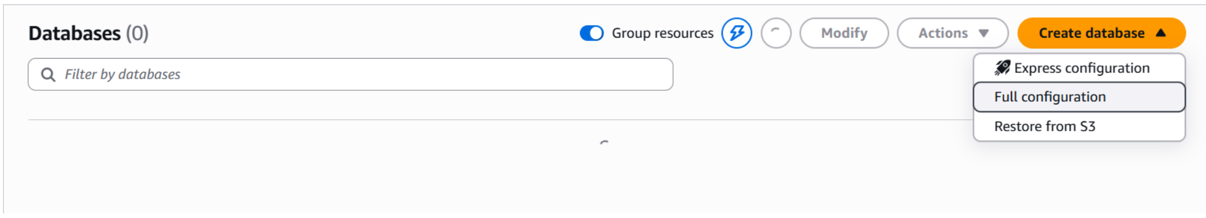
Amazon RDS for MySQL Zero-ETL Integration with Amazon Redshift

To Begin with the Lab

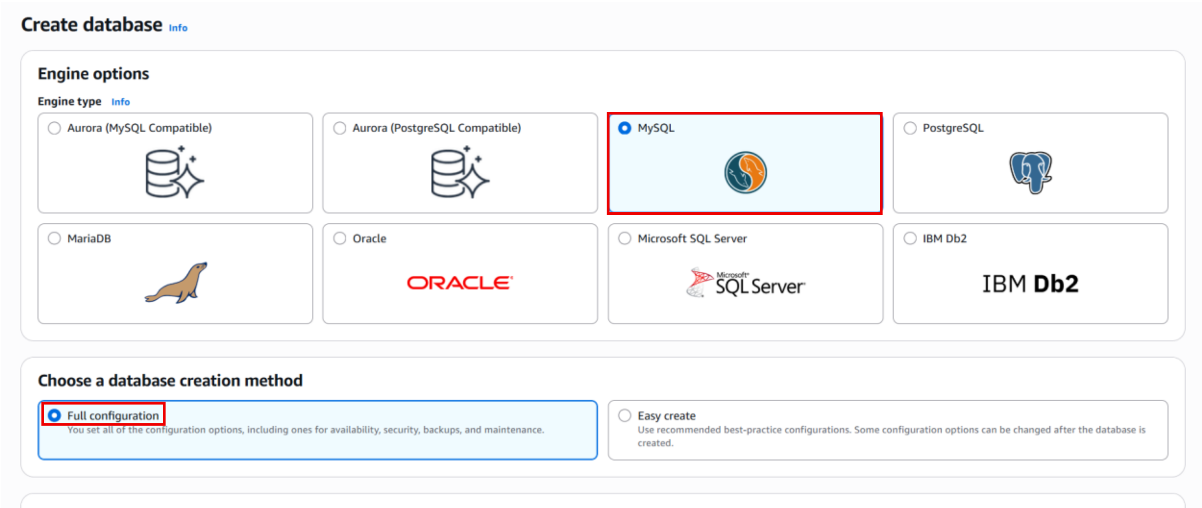
Summary of the Lab

This lab demonstrates how to configure an **Amazon RDS for MySQL Zero-ETL integration with Amazon Redshift Serverless**. First, an RDS MySQL instance is created, connected using MySQL Workbench, and populated with a sample database containing employee and department tables. Next, an Amazon Redshift Serverless workgroup and namespace are created. A Zero-ETL integration is then configured between the RDS instance and the Redshift namespace, with AWS automatically applying the required settings. Although the integration becomes active, the destination database is not created automatically, preventing tables from appearing in Redshift. To resolve this, a destination database is manually created from the integration page in Redshift. After the database is provisioned and activated, the synchronized MySQL database and tables become available in the Redshift Query Editor, enabling near real-time analytics without building ETL pipelines.

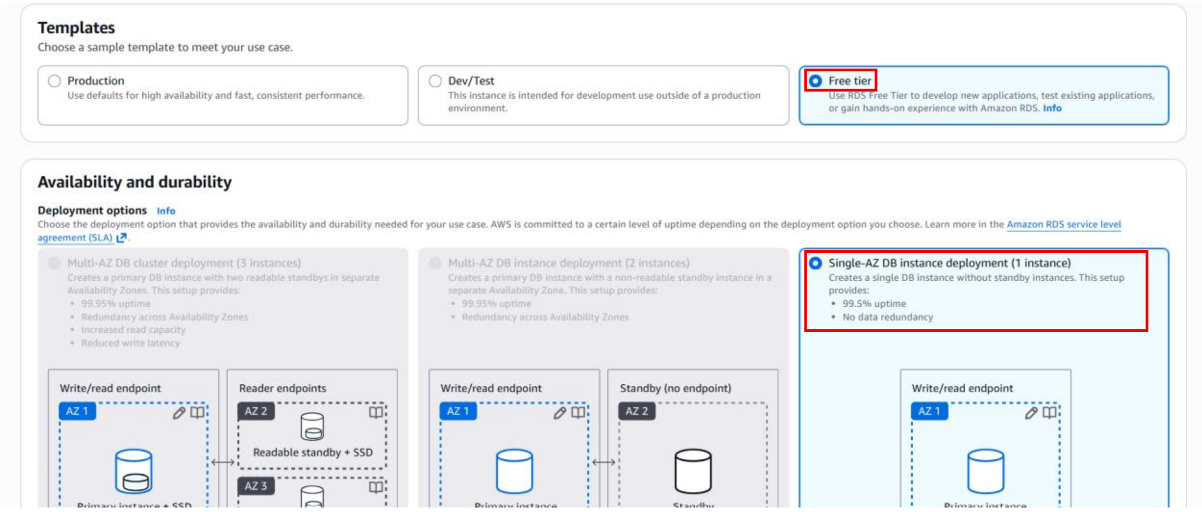
- Open Amazon RDS
- Login to AWS Console
- Search for RDS
- Open Amazon RDS
- Click Create Database
- Choose Full Configuration



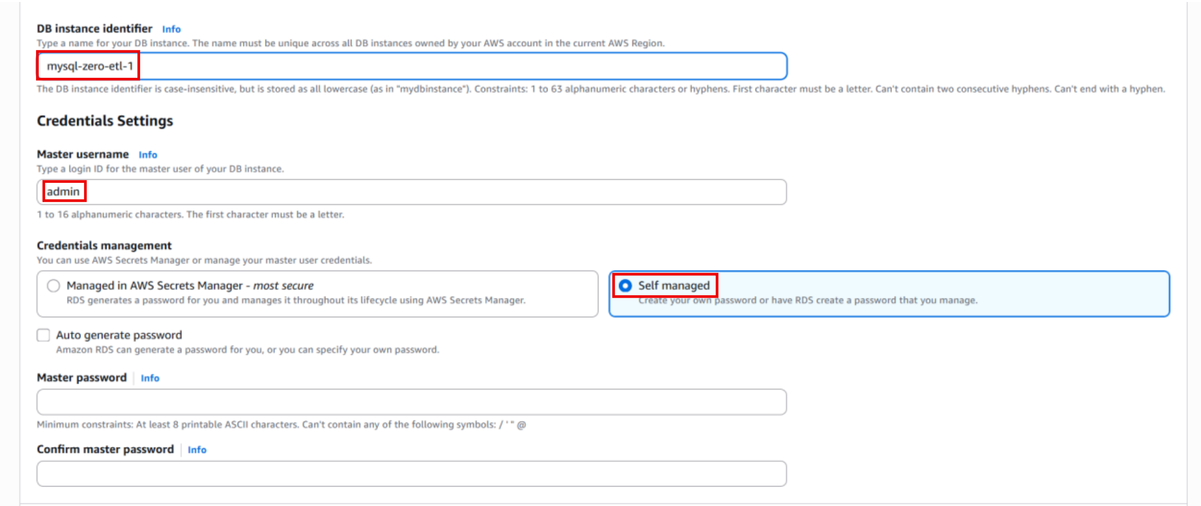
- Choose Database Creation Method
- Select:
- Standard Create
- Select Engine
- Choose:
- Engine Type: MySQL
- Version:
- MySQL 8.0.x
- (Use the latest supported version)



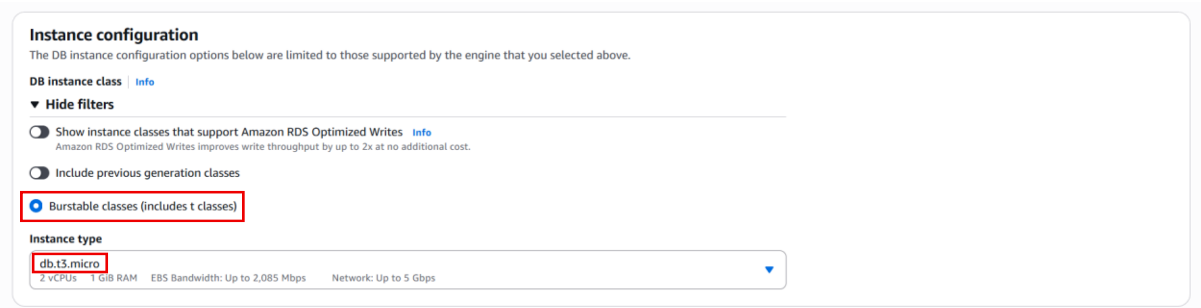
- Select Template
- Choose:
- Free Tier
- (or Dev/Test if Free Tier is unavailable)



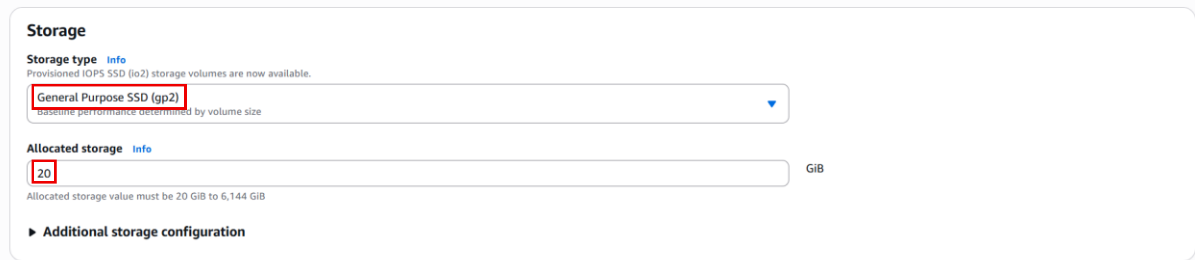
- Configure DB Instance
- DB Instance Identifier
    - mysql-zero-etl
- Master Username
    - Admin
- Choose Self-Managed
- Master Password
- Confirm Password



- Instance Configuration
- Choose Burstable classes
- Choose:
    - db.t3.micro
- (or any supported instance class)

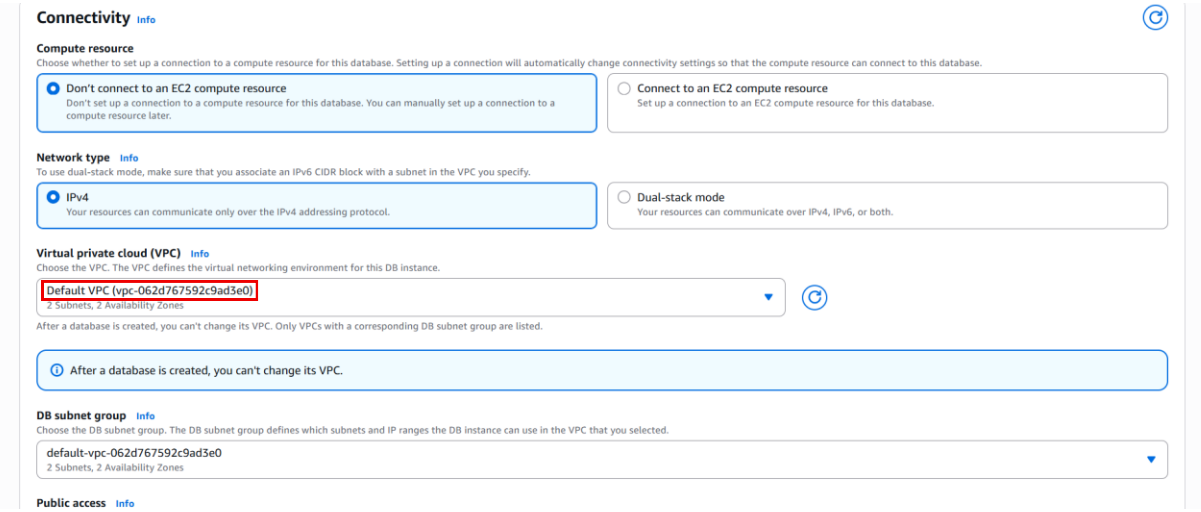


- Storage Configuration
- Storage Type
    - General Purpose SSD (gp3)
- Allocated Storage
    - 20 GB

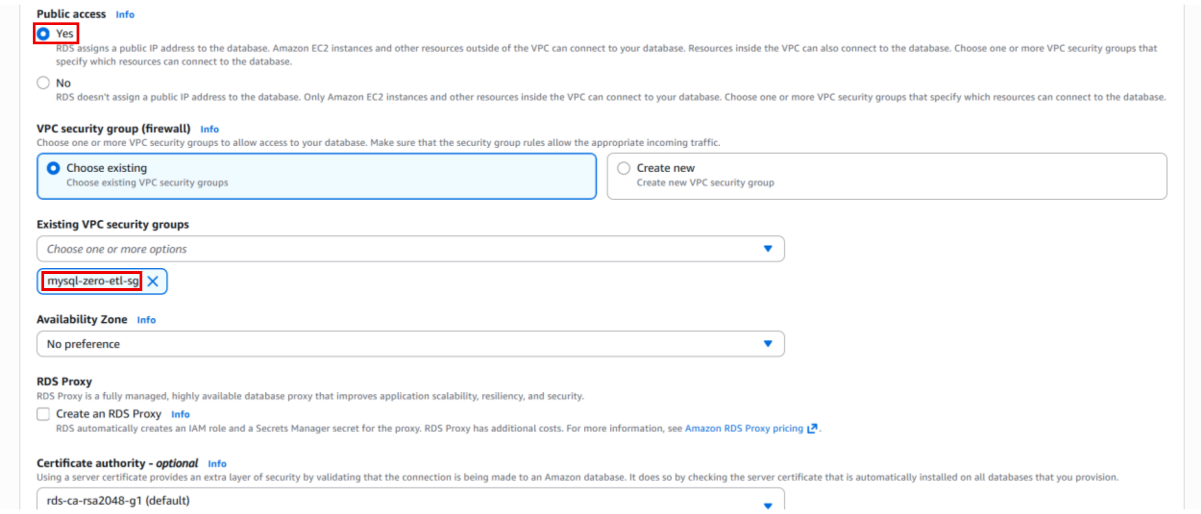


- Connectivity
- VPC
- Select:
- Default VPC

- or your custom VPC.

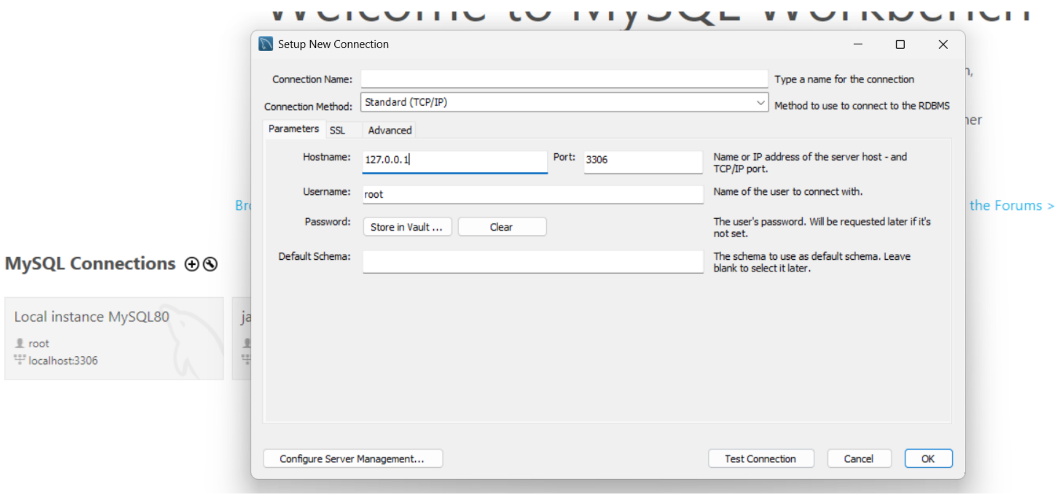


- Public Access
- For MySQL Workbench connection:
- Yes
- VPC Security Group
- Create new:
    - mysql-zero-etl-sg
- or use existing security group.
- Database Port
    - 3306

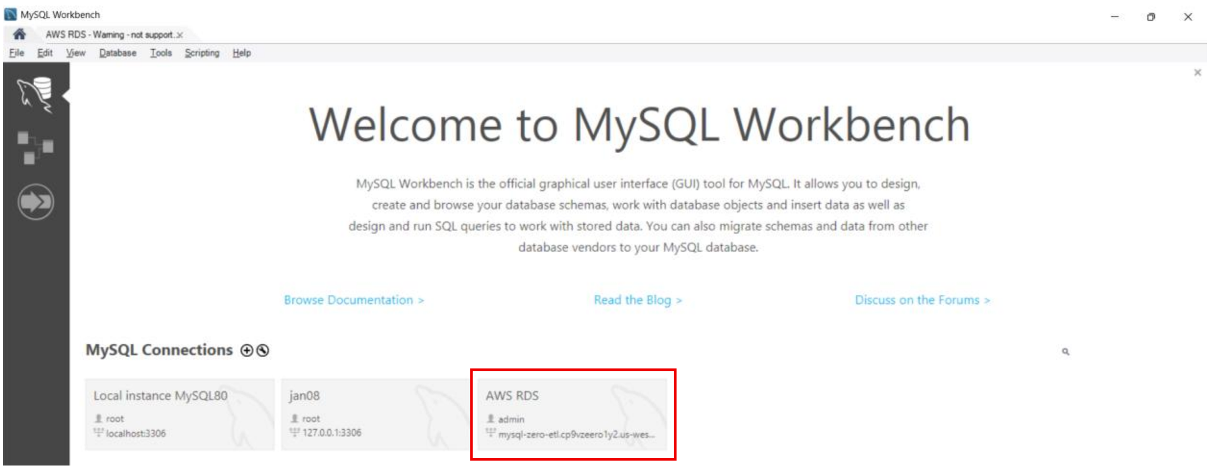


- Click:
    - Create Database
- Wait for:
- Status = Available
- Connect Using MySQL Workbench
- Use:
    - Hostname = RDS Endpoint

- Port = 3306
- Username = admin
- Password = Your Password



- Test Connection → OK



- Create Sample Database
- Run:

```
CREATE DATABASE companydb;

USE companydb;
```

- Step 4: Create Tables
- Employees Table

```
CREATE TABLE employees (

    employee_id INT PRIMARY KEY,

    employee_name VARCHAR(100),
```

```
    department VARCHAR(50),
    salary DECIMAL(10,2)
);
```

- Departments Table

```
CREATE TABLE departments (
    department_id INT PRIMARY KEY,
    department_name VARCHAR(100)
);
```

- Step 5: Insert Sample Data
- Employees

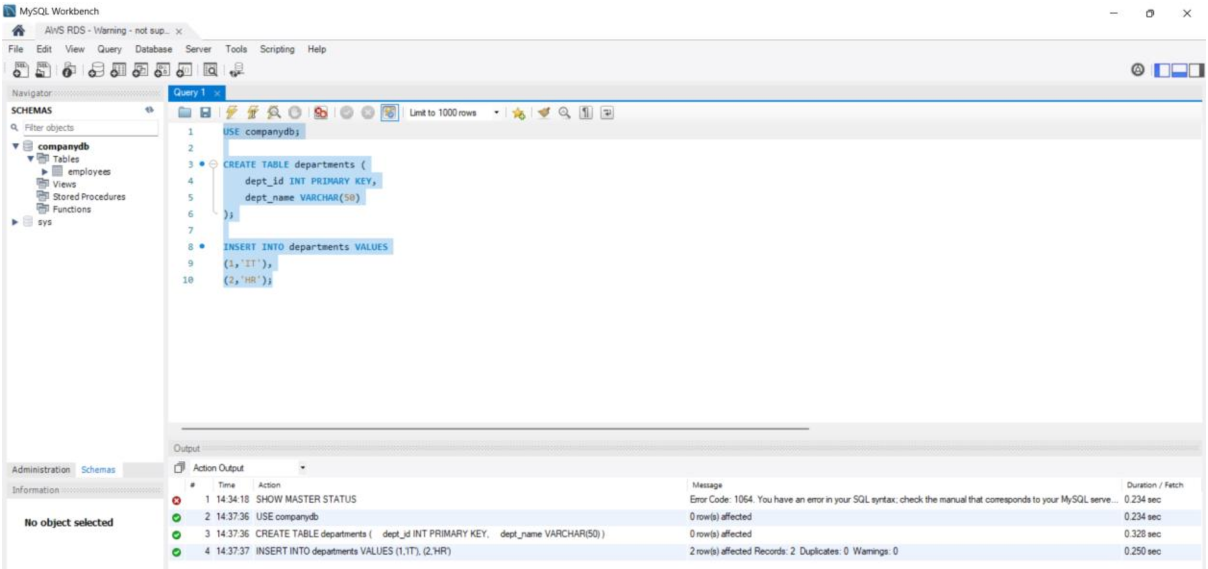
```
INSERT INTO employees
VALUES
(101,'Aman','IT',50000),
(102,'Rahul','HR',45000),
(103,'Priya','Finance',60000);
```

- Departments

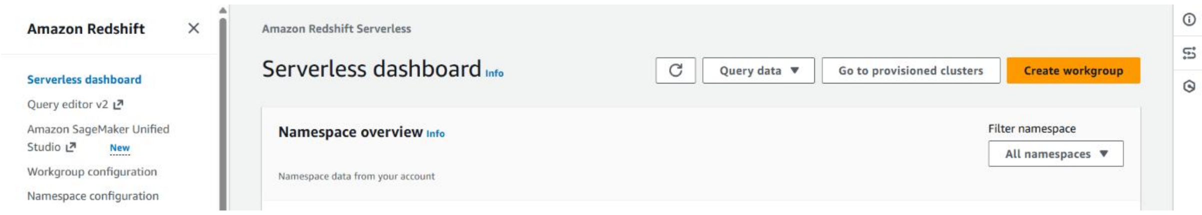
```
INSERT INTO departments
VALUES
(1,'IT'),
(2,'HR'),
(3,'Finance');
```

- Verify:

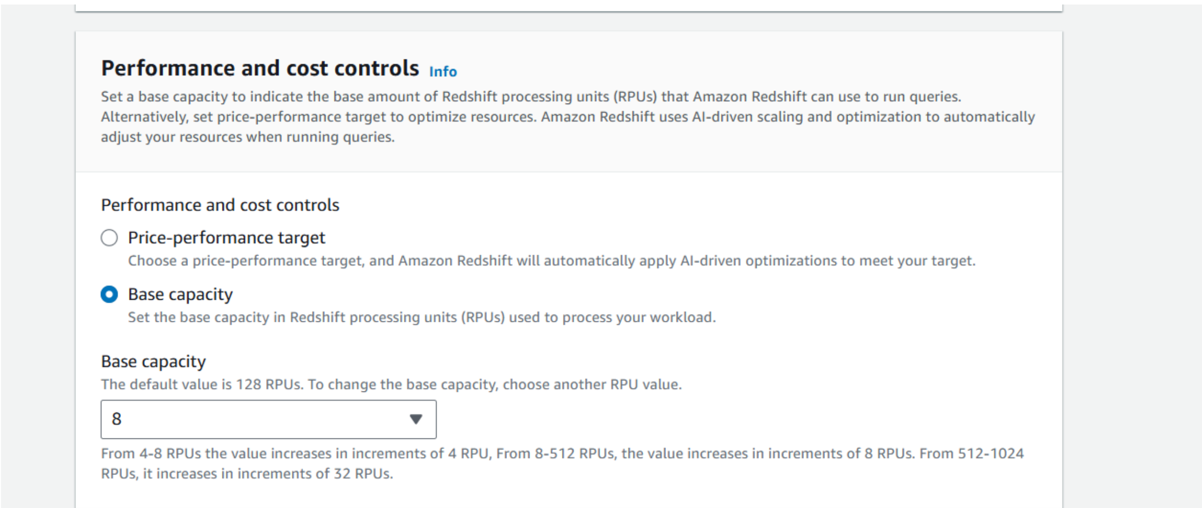
```
SELECT * FROM employees;
SELECT * FROM departments;
```



- Create Redshift Serverless
- Open:
- AWS Console → Amazon Redshift → Serverless → Create Workgroup

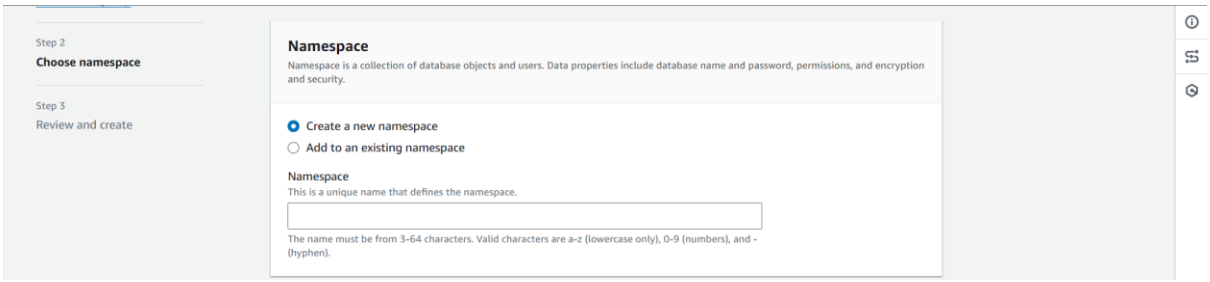


- Configure Workgroup
- Workgroup Name
    - zero-etl-workgroup
- Base Capacity
    - Selected:
        - 8 RPUs

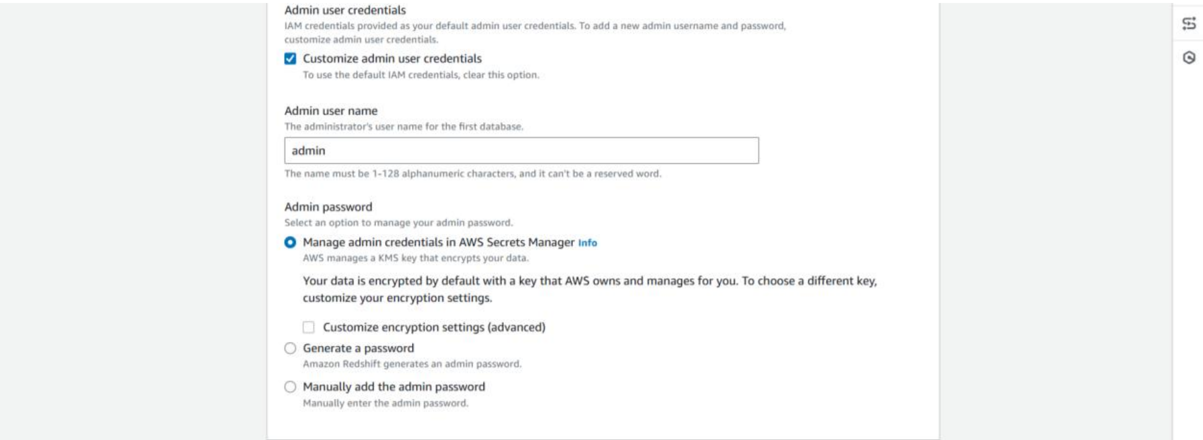


- Configure Namespace

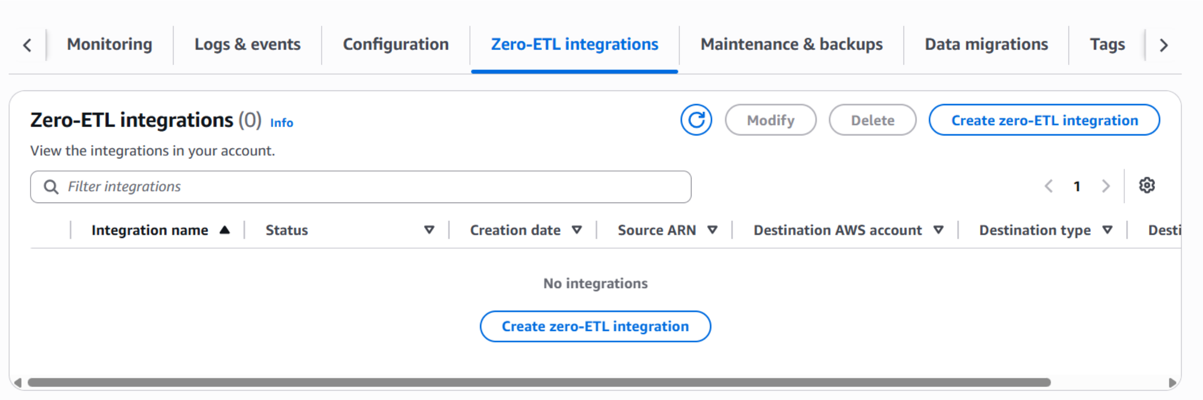
- Namespace Name
  - zero-etl-namespace



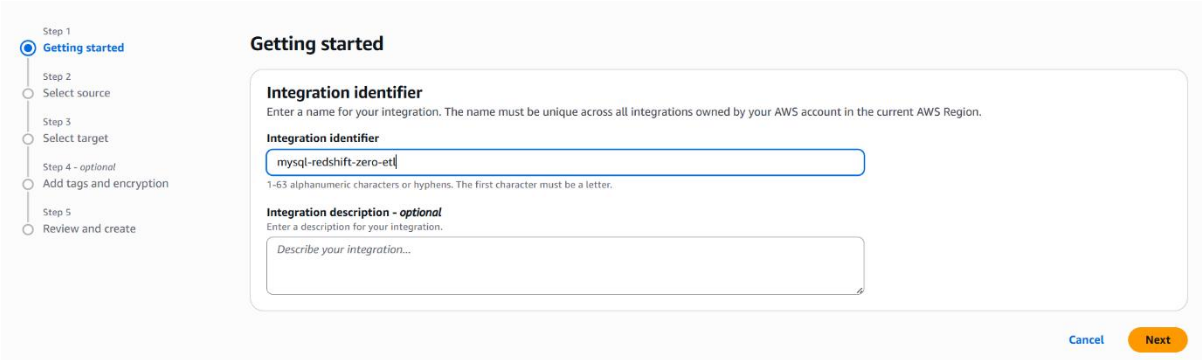
- Admin Username
  - admin
- Admin Password
  - <Your Password>
- Database Name
  - dev
- Create.
- Wait until:
- Status = Available



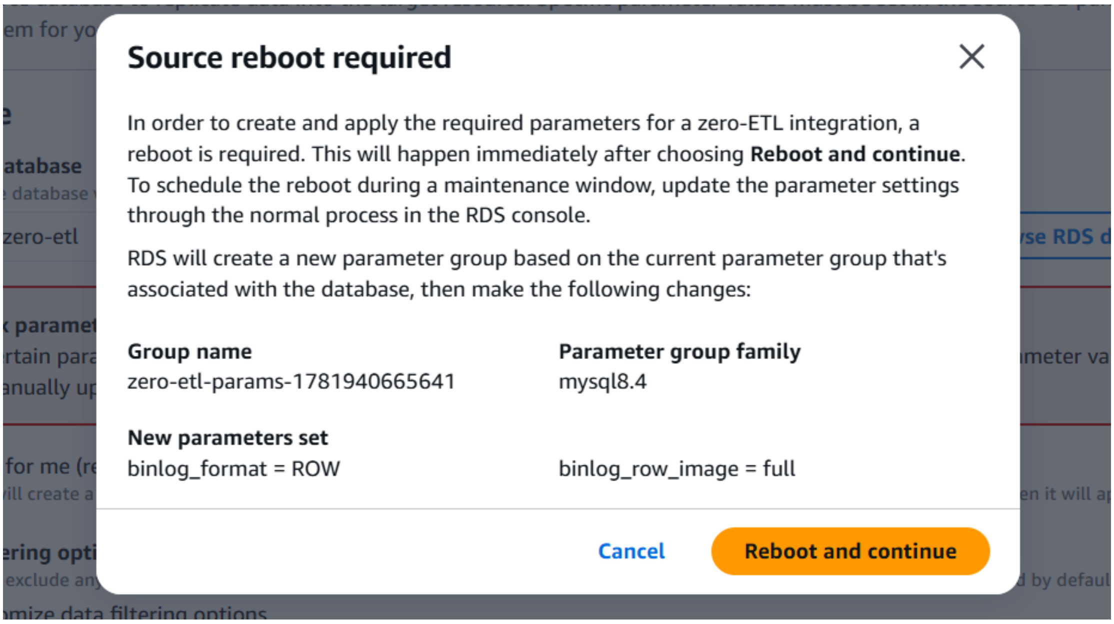
- Create Zero-ETL Integration
- Open:
- mysql-zero-etl → Zero-ETL Integrations → Create Integration



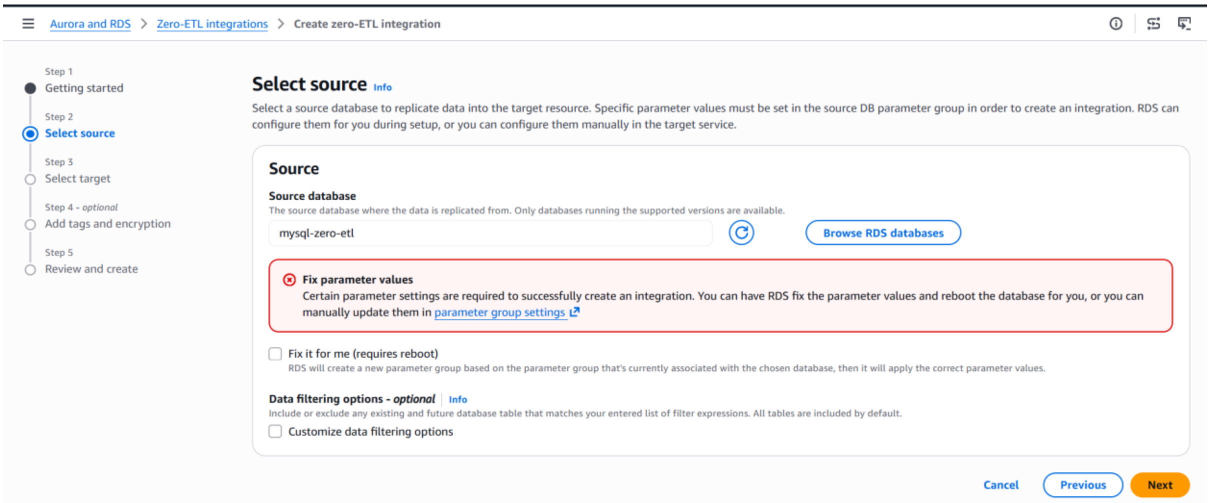
- Integration Name
    - mysql-redshift-zero-etl



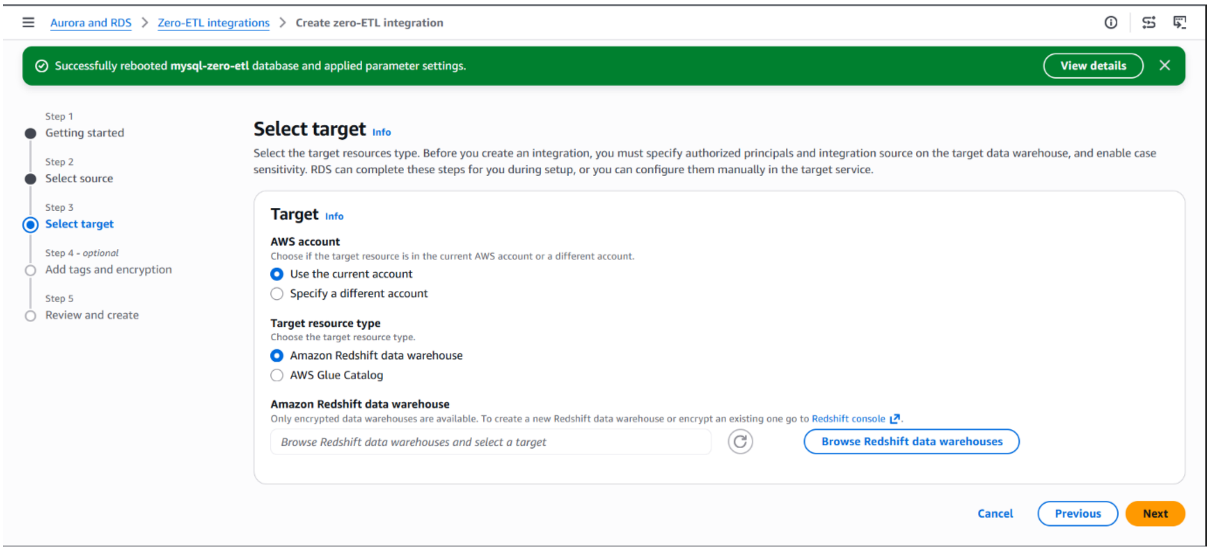
- Source
    - mysql-zero-etl
- Check the Fix it for me
- Click Reboot and continue



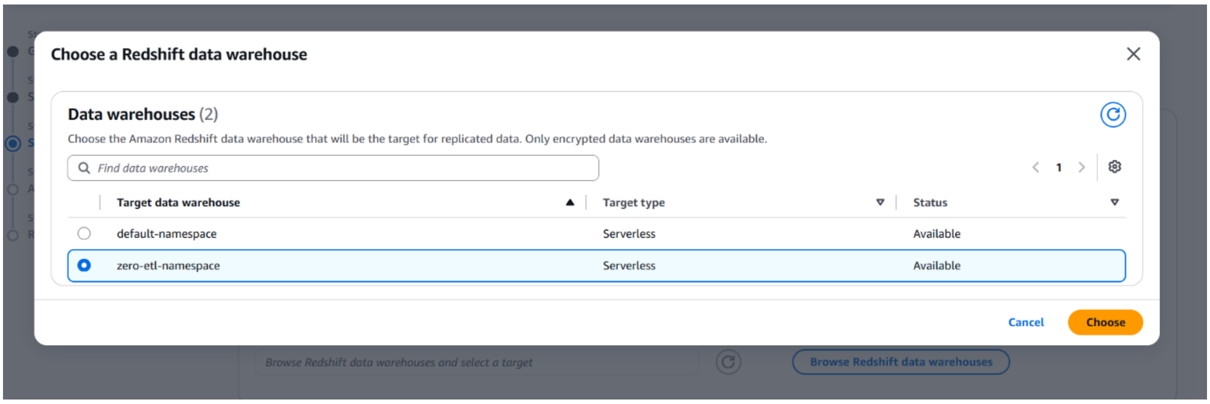
- Click Next



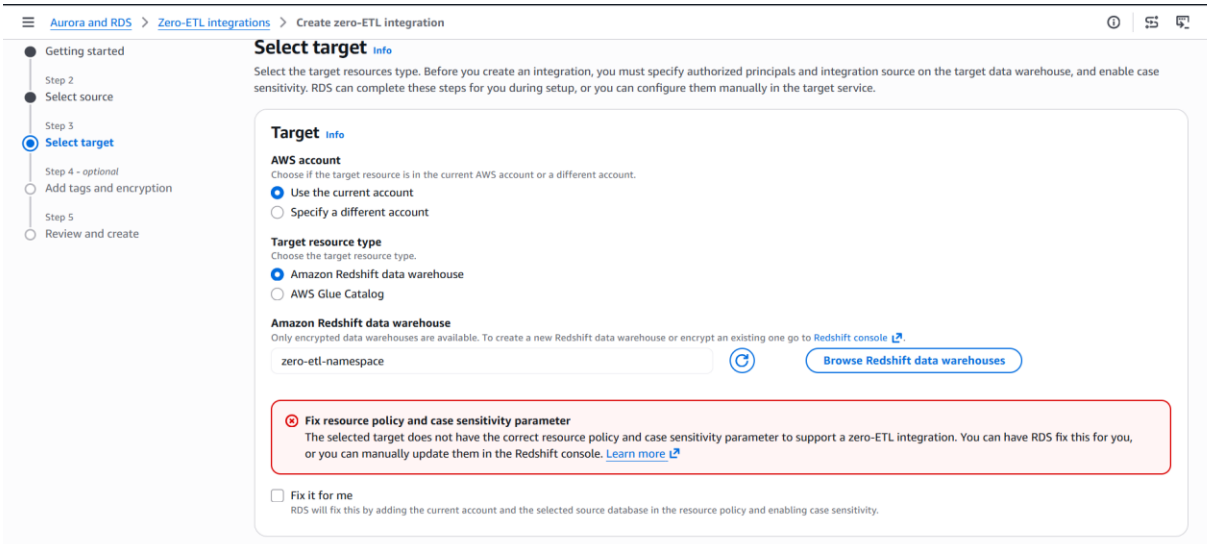
- Target
- Amazon Redshift Data warehouse



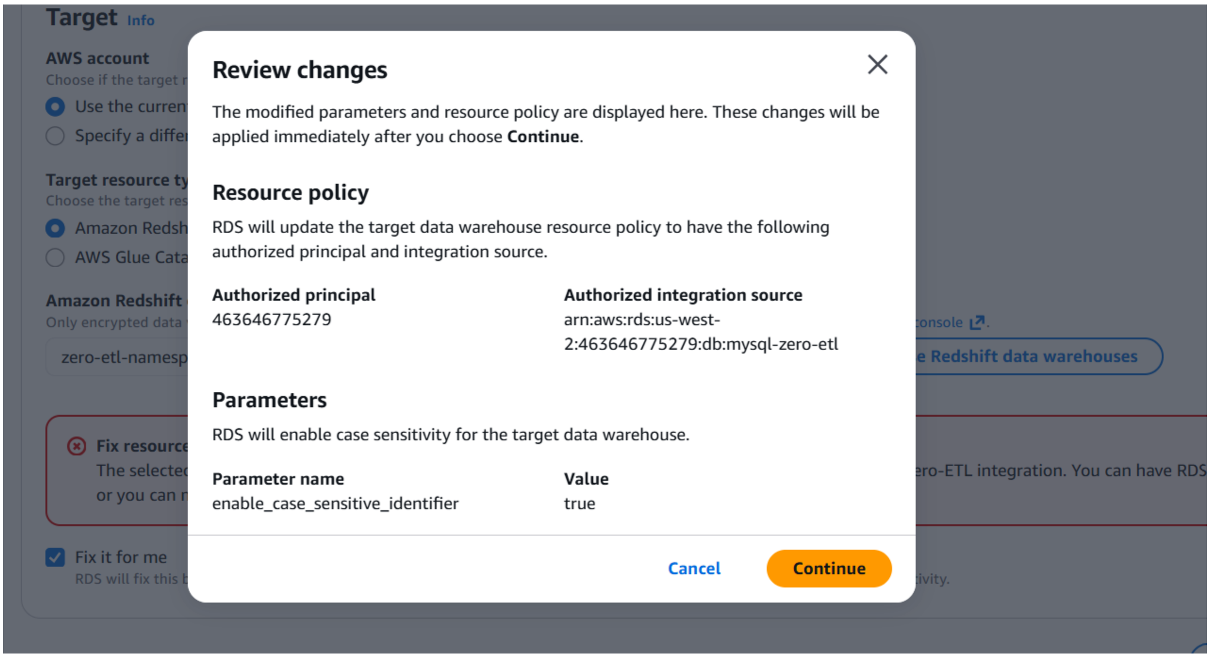
- Select:
- Namespace:
    - zero-etl-namespace



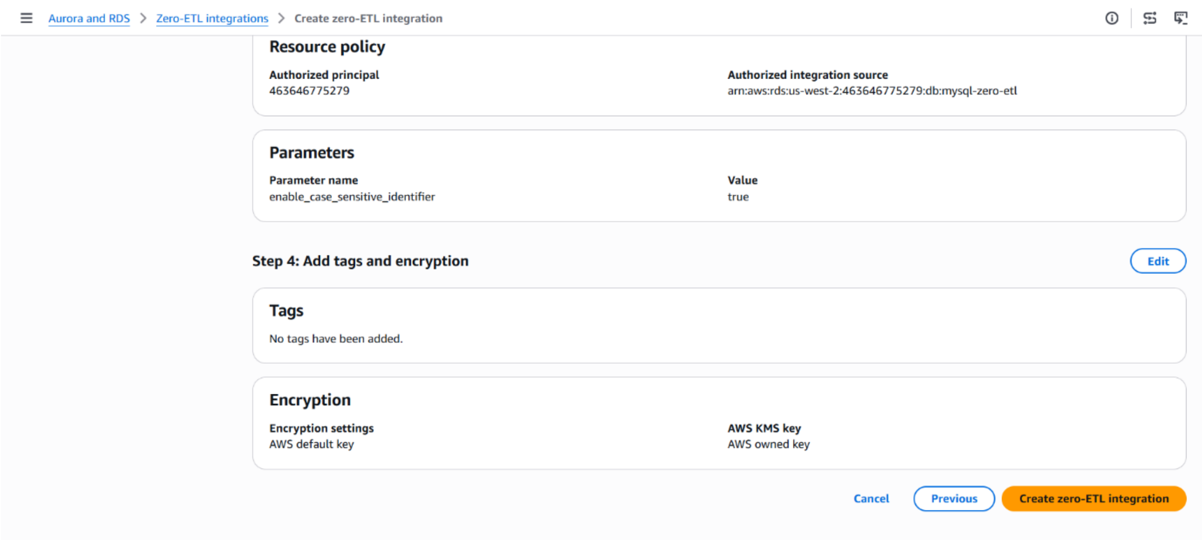
- Check Fix it for me



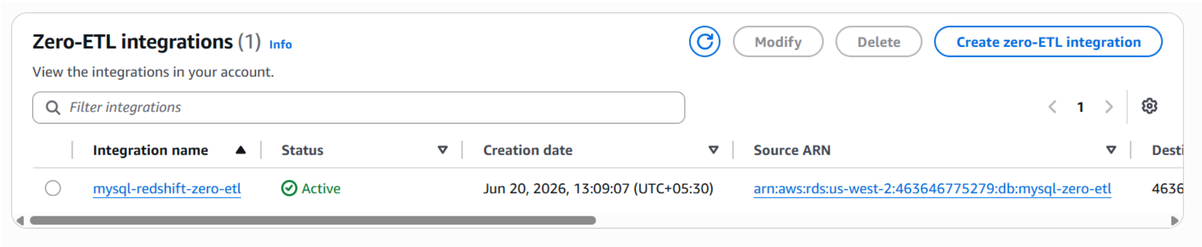
- Click Continue



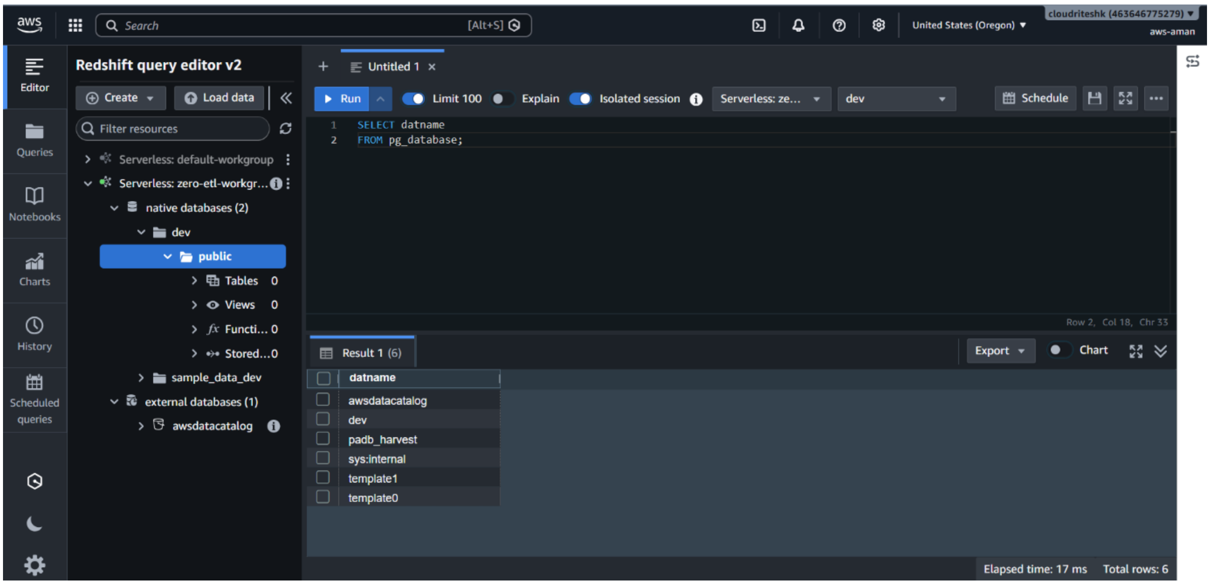
- Create Integration.



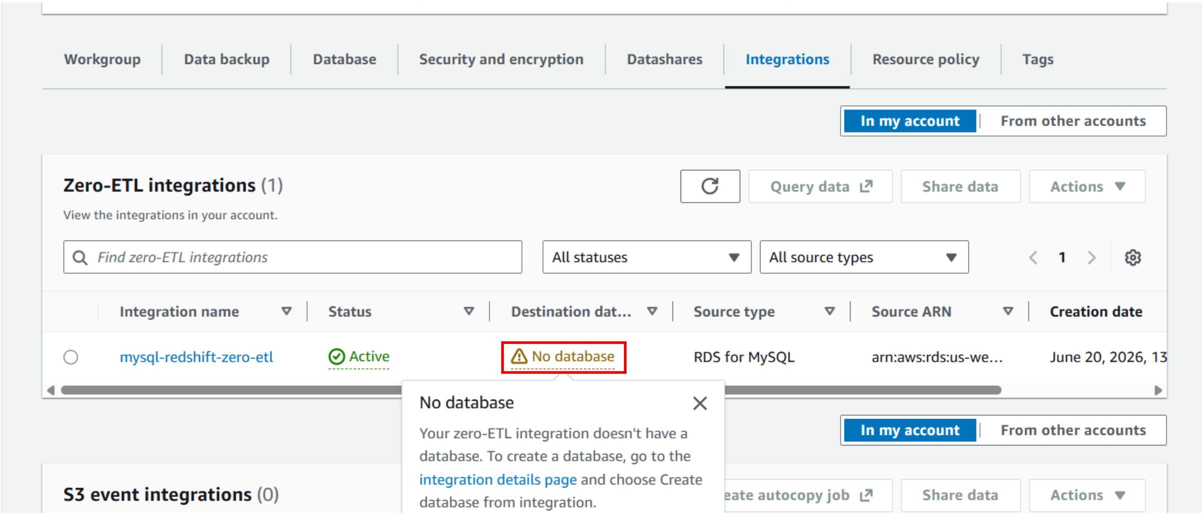
- Wait For Integration
- Initially:
  - Status = Creating
- Wait until:
  - Status = Active

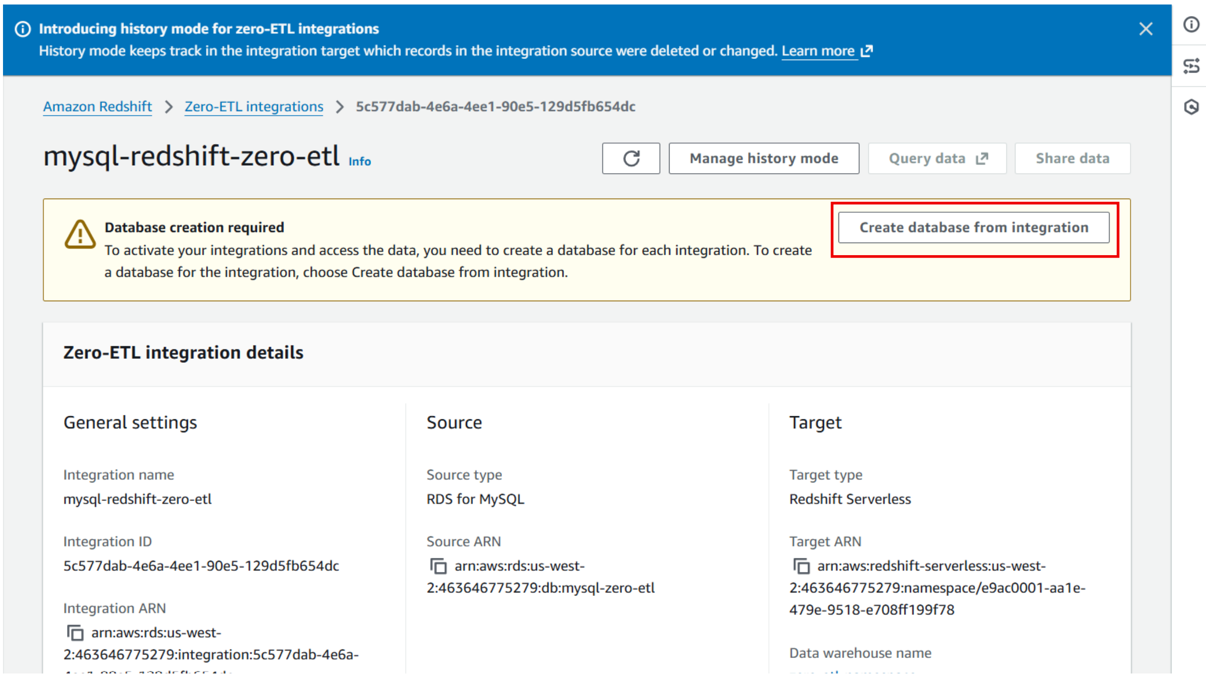


- First Problem We Faced
- Integration became:
  - Active
- But in Redshift:
  - No tables
  - No schemas
  - No database

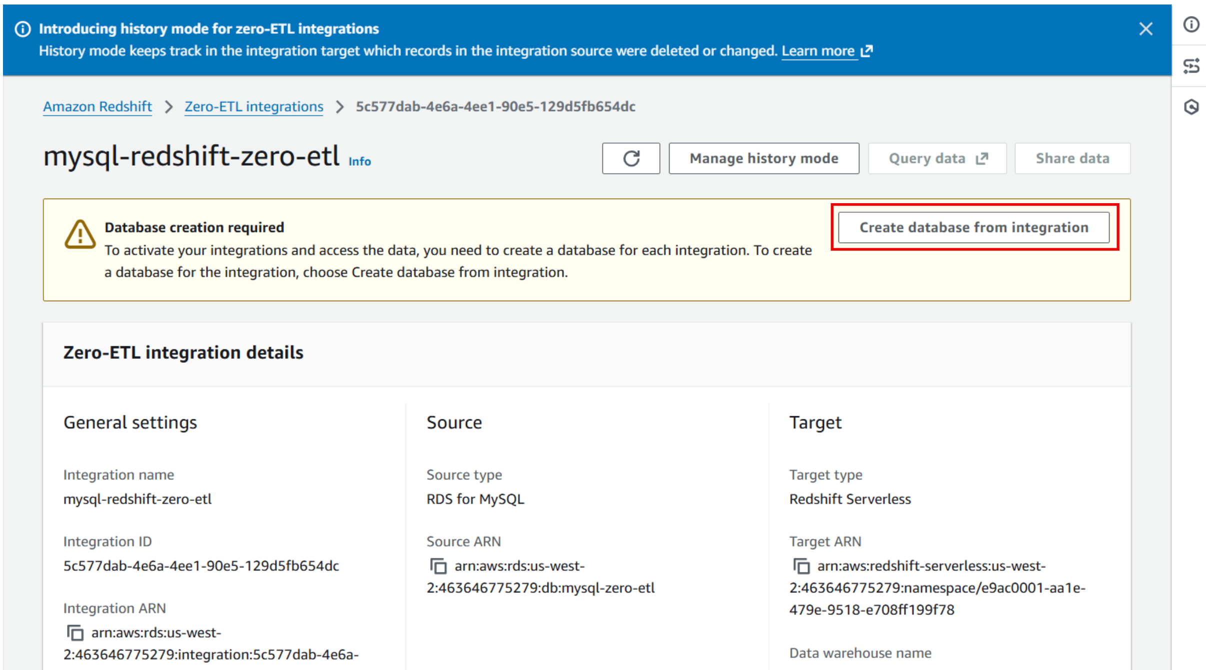


- Root Cause
- Integration was active but destination database was not created.
- Integration page showed:
- Destination database:
    - No database
- and
- Database creation required



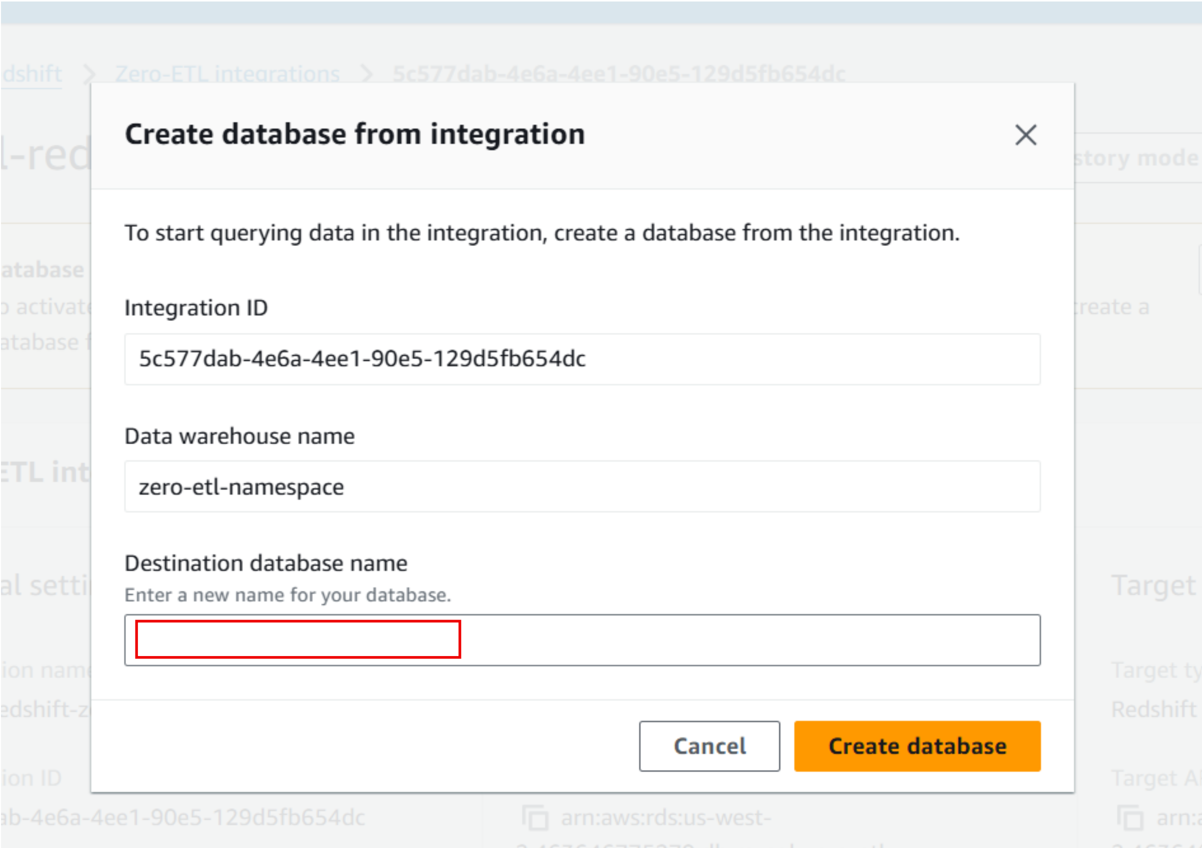


- To fix this, Database creation required
- Create Database from Integration
- Open:
- Redshift → Zero-ETL Integrations → mysql-redshift-zero-etl
- Click:
- Create database from integration

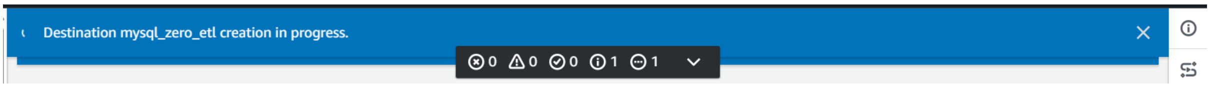


- Enter:
- Destination Database Name:
  - **mysql_zero_etl**
- Click:

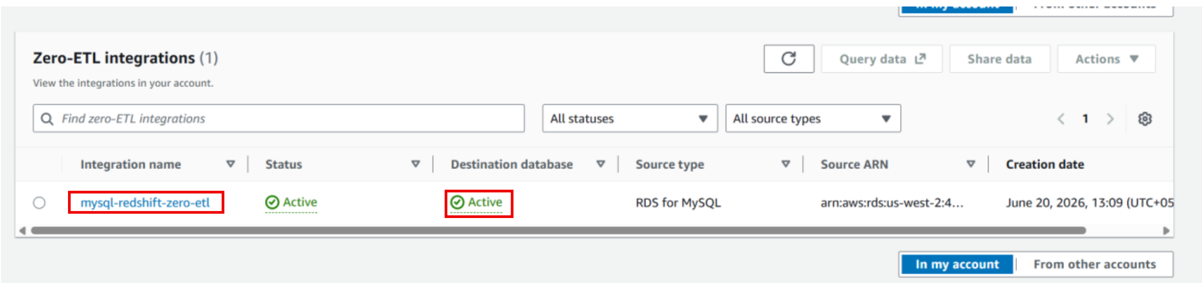
- Create Database



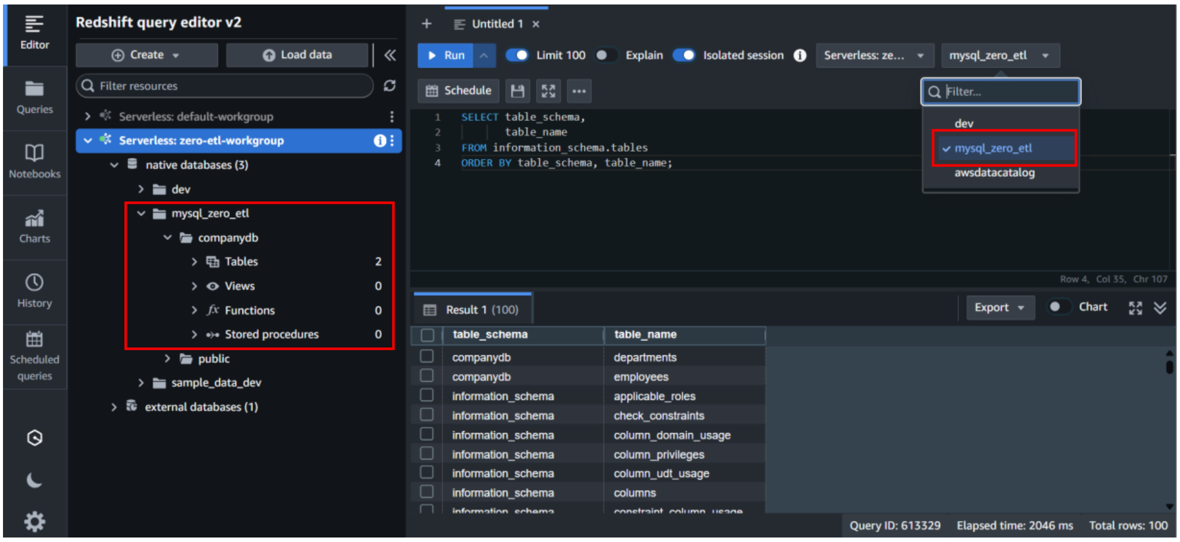
- Started creating, it will take 10 to 15 min.



Once it is done, Destination database will be activated.



- Now go back to query editor, you can see the database there with the tables.


Redshift query editor v2
Create
Load data
Filter resources
Serverless: default-workgroup
Serverless: zero-etl-workgroup
native databases (3)
dev
mysql_zero_etl
companydb
Tables 2
Views 0
Functions 0
Stored procedures 0
public
sample_data_dev
external databases (1)
Untitled 1
Run
Limit 100
Explain
Isolated session
Serverless: ze...
mysql_zero_etl
Schedule
SELECT table_schema,
table_name
FROM information_schema.tables
ORDER BY table_schema, table_name;
Filter...
dev
mysql_zero_etl
awsdatacatalog
Result 1 (100)
Export
Chart
table_schema
table_name
companydb departments
companydb employees
information_schema applicable_roles
information_schema check_constraints
information_schema column_domain_usage
information_schema column_privileges
information_schema column_udt_usage
information_schema columns
Query ID: 613329
Elapsed time: 2046 ms
Total rows: 100
Editor
Queries
Notebooks
Charts
History
Scheduled queries